THE MASTERMIND COACHING CLASSES

CBSE / RBSE · Nursery – Class XII · Road No. 5, VKIA, Jaipur · 9887258679

CLASS IX — MATHEMATICS · FULL SYLLABUS (PART 1 + PART 2)

Chapters 1–14 | Session 2026-27 | CBSE / RBSE Pattern

Class: 9	Subject: Maths	Set 01	Time: 3 Hours	Max. Marks: 80

General Instructions:

1. This question paper contains 38 questions and is divided into 5 sections — A, B, C, D and E.
2. In Section A, Q. 1–18 are multiple-choice questions and Q. 19–20 are Assertion–Reason based questions of 1 mark each.
3. In Section B, Q. 21–25 are very short answer (VSA) type questions carrying 2 marks each.
4. In Section C, Q. 26–31 are short answer (SA) type questions carrying 3 marks each.
5. In Section D, Q. 32–35 are long answer (LA) type questions carrying 5 marks each.
6. In Section E, Q. 36–38 are case-study based questions carrying 4 marks each with sub-parts of 1, 1 and 2 marks respectively.
7. All questions are compulsory. However, an internal choice is provided in 1 question of Section B, 2 questions of Section C and 2 questions of Section D. An internal choice is provided in all the 2-mark sub-parts of Section E.
8. Draw neat and clean figures wherever required. Take π = 22/7 wherever required, if not stated. Use of calculators is not allowed.
9. Write the serial number of each question before attempting it.

SECTION A — Objective Type — 20 questions · 20 marks

Q1–Q18 are multiple-choice questions (1 mark each). Q19–Q20 are Assertion (A) and Reason (R) questions: select the correct option from (a), (b), (c), (d) given below each.

1. The point (0, −7) lies on the: **1**

(a) fourth quadrant (b) third quadrant
(c) x-axis (d) y-axis

2. The distance between the points A(−2, 3) and B(4, −5) is: **1**

(a) √28 units (b) 8 units (c) 10 units (d) 14 units

3. Which of the following is a solution of the linear equation 5x + 2y = 20? **1**

(a) (2, 5) (b) (3, 2) (c) (5, 2) (d) (4, 1)

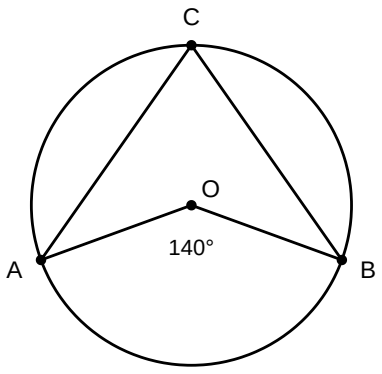
4. Which of the following is an irrational number? **1**

(a) 0.2525… (25 repeating) (b) √(49/16)
(c) √0.09 (d) √50

5. The product $(x + 4)(x - 4)(x^2 + 16)$ is equal to: **1**

(a) $x^4 - 8x^2 + 256$ (b) $x^4 - 256$ (c) $x^4 + 256$ (d) $x^4 - 16$

6. In the figure, O is the centre of the circle and ∠AOB = 140°. Then ∠ACB is equal to: **1**



(a) 35° (b) 140° (c) 70° (d) 280°

7. A chord of length 48 cm is drawn in a circle of radius 25 cm. Its distance from the centre is: **1**

(a) 24 cm (b) 14 cm (c) 7 cm (d) √1201 cm

8. The sides of a triangle are in the ratio 3 : 5 : 7 and its perimeter is 60 cm. The longest side is: **1**

(a) 12 cm (b) 20 cm (c) 35 cm (d) 28 cm

9. A letter is chosen at random from the word 'MATHEMATICS'. The probability that it is a vowel is: **1**

(a) $\frac{4}{11}$ (b) $\frac{5}{11}$ (c) $\frac{7}{11}$ (d) $\frac{1}{4}$

10. If the n-th term of a sequence is $t_n = 3n^2 - 2$, then t_5 is: **1**

(a) 77 (b) 73 (c) 28 (d) 13

11. The converse of the proposition 'If a number is divisible by 15, then it is divisible by 5' is: **1**

(a) If a number is not divisible by 15, then it is not divisible by 5.

(b) If a number is divisible by 5, then it is divisible by 15.

(c) If a number is divisible by 5, then it is divisible by 3.

(d) A number is divisible by 15 only if it is divisible by 5.

12. A counterexample to the claim 'If n is a prime number, then $2^n - 1$ is a prime number' is: **1**

(a) n = 2 (b) n = 7 (c) n = 5 (d) n = 11

13. In a class, 20 students have an average mark of 60 and the other 30 students have an average mark of 70. The average mark of all 50 students is: **1**

(a) 66 (b) 67 (c) 64 (d) 65

14. The volume of a cone of base radius 6 cm and height 7 cm (take $\pi = 22/7$) is: **1**

(a) 792 cm^3 (b) 528 cm^3 (c) 264 cm^3 (d) 132 cm^3

15. When the algorithm that checks every j from 1 to n is executed for n = 20, the final list of divisors is: **1**

(a) [1, 2, 4, 5, 10, 15, 20] (b) [2, 4, 5, 10]

(c) [1, 2, 4, 5, 10, 20] (d) [1, 2, 5, 10, 20]

16. In a parallelogram ABCD, $\angle A = 72°$. Then $\angle B$ is equal to: **1**

(a) 118° (b) 108° (c) 98° (d) 72°

17. In $\triangle ABC$, D and E are the midpoints of AB and AC respectively. If BC = 18 cm, then DE is: **1**

(a) 36 cm (b) 6 cm (c) 9 cm (d) 18 cm

18. In a sequence $t_1 = 5$ and $t_n = t_{n-1} + 3$ for $n \geq 2$. The sixth term is: **1**

(a) 17 (b) 23 (c) 18 (d) 20

19. **Assertion (A):** $\sqrt{3} \times \sqrt{12}$ is a rational number. **1**
Reason (R): The product of two irrational numbers is always irrational.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

20. **Assertion (A):** A quadrilateral whose diagonals are equal must be a rectangle. **1**
Reason (R): The diagonals of a rectangle are equal.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION B — Very Short Answer (2 marks each) — 5 questions · 10 marks

Internal choice is provided in one question.

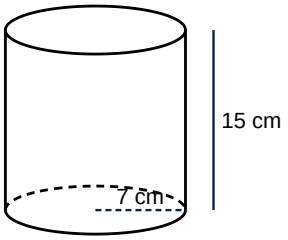
21. A gym charges ₹800 as joining fee and ₹350 per month. Write an expression for the total amount paid for m months and find the amount for 6 months. **2**

22. Without actual division, decide whether $\frac{29}{200}$ has a terminating decimal expansion. If it does, find the number of decimal places. **2**

OR

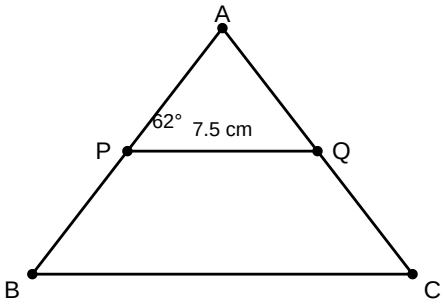
Find two rational numbers between $\frac{1}{6}$ and $\frac{1}{5}$.

23. A cylindrical tin of radius 7 cm and height 15 cm is open at the top. Find the area of tin sheet needed to make it (π = 22/7). **2**



24. Write the converse of the proposition 'If a number is divisible by 12, then it is divisible by 6.' Is the converse true? Justify. **2**

25. In the figure, P and Q are the midpoints of AB and AC and PQ = 7.5 cm, ∠APQ = 62°. Find BC and ∠ABC, giving reasons. **2**



SECTION C — Short Answer (3 marks each) — 6 questions · 18 marks

Internal choice is provided in two questions.

26. Show that 0.538461538461… = $\frac{7}{13}$. **3**

27. The mode of travel to school of 100 students was recorded: **3**

Mode	Walk	Cycle	Bus	Car
Students	24	36	28	12

A student is chosen at random. Find the probability that the student (i) cycles, (ii) walks or comes by bus, (iii) does not come by car.

OR

The number of books read last month by 80 students was recorded:

Books	0	1	2	3
Students	12	28	24	16

A student is chosen at random. Find the probability that the student read (i) exactly 2 books, (ii) at least 2 books, (iii) fewer than 2 books.

28. Draw the graphs of the equations x + y = 6 and x − y = 2 on the same grid and use them to find the solution of the pair. **3**

29. Meera's marks are: unit tests 80%, project 70%, final examination 90%. The annual score is found by combining the three in the ratio 2 : 3 : 5. Find her annual score and say why a simple average of 80, 70 and 90 would not be correct. **3**

30. Using Heron's formula, find the area of a triangle whose sides are 9 cm, 10 cm and 17 cm. **3**

OR

The sides of a triangle are in the ratio 9 : 12 : 15 and its perimeter is 108 cm. Find its area.

31. Execute the final gcd algorithm of the chapter (start with most-recent-common-divisor = 1 and check every k from 2 to min(m, n)) for m = 9 and n = 12. Show each step and write the gcd. **3**

SECTION D — Long Answer (5 marks each) — 4 questions · 20 marks

Internal choice is provided in two questions.

32. A craftsman marks the corners of a metal plate on a grid as P(1, 1), Q(5, 1), R(7, 4) and S(3, 4). **5**
(a) Plot the points and join them in order. (2 marks)
(b) Find the lengths PQ, QR, RS and SP. (1½ marks)
(c) Show that PQRS is a parallelogram by finding the midpoints of the diagonals PR and QS. (1½ marks)

33. How many three-digit numbers are multiples of 7? Find their sum. (2½ + 2½) **5**

OR

A tyre company tested 800 tyres and recorded the distance they ran:

Distance (thousand km)	0 – 20	20 – 40	40 – 60	60 – 80
Number of tyres	40	120	360	280

Find the probability that a tyre chosen at random runs (i) at least 40 thousand km (1), (ii) less than 20 thousand km (1), (iii) between 20 and 60 thousand km (2), and (iv) state the sum of all four class probabilities (1).

34. State and prove the Midpoint Theorem: the segment joining the midpoints of two sides of a triangle is parallel to the third side and equal to half of it. **5**

OR

Batch A of brass weighs 150 kg and is 60% copper. Batch B weighs 100 kg and is 40% copper. Batch C is 45% copper. All three batches are melted together and the alloy is found to be 50% copper. Find the weight of Batch C.

35. The length of a rectangular plot is 4 m more than its breadth and its area is 96 m². Form a quadratic expression in the breadth x, factorise it and find the perimeter of the plot. **5**

OR

A carpenter makes a rectangle from 2 square tiles of side x units, 11 rectangular strips of size x by 1 unit and 12 unit squares. Write the total area as a polynomial, factorise it and find the dimensions of the rectangle.

SECTION E — Case-Study Based Questions (4 marks each) — 3 questions · 12 marks

Read each case carefully and answer the questions that follow. Internal choice is provided in every 2-mark sub-part.

36. Ananya saves for a trip. In week 1 she saves ₹150 and every week she saves ₹25 more than in the week before. The amount saved in week n is given by $t_n = 125 + 25n$.

(i) How much does she save in week 12? 1

(ii) What type of sequence is this and what is the common difference? 1

(iii) In which week does her weekly saving first reach ₹500? 2

OR

(iii) Find the total amount she saves in the first 10 weeks.

37. A triangular plot of land has sides 25 m, 39 m and 56 m.

(i) Find the semi-perimeter of the plot. 1

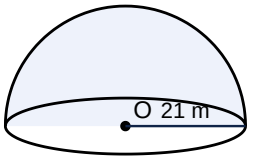
(ii) Find its area using Heron's formula. 1

(iii) Find the length of the perpendicular from the opposite vertex to the longest side. 2

OR

(iii) The owner fences the plot at ₹80 per metre and levels the ground at ₹12 per m². Find the cost of fencing and the cost of levelling.

38. The dome of a planetarium is a hemisphere of radius 21 m (take π = 22/7).



Dome

(i) Find the curved surface area of the dome. 1

(ii) Find the volume of air inside the dome. 1

(iii) The outside of the dome is painted at ₹5 per m². Find the cost. 2

OR

(iii) A cylinder has the same radius and height 21 m. Show that volume of dome : volume of cylinder = 2 : 3.